



Secure Exit

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT3-Debugging and Optimization

View Only

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QUESTIONS

Q1

ATM Withdrawal – Exception Handling Debugging

10 marks

Q2

Hospital Registration – NullPointerException Debugging

10 marks

Q3

E-Commerce Product Search – Array Exception Debugging

10 marks

Q4

Bank Account – Race Condition Debugging

10 marks

Q5

Railway Reservation – Synchronization Debugging

Q2 Hospital Registration – NullPointerException Debugging

10 marks

A hospital patient registration application crashes when patient information is unavailable.

Debug the following complete Java program.

Task:

1. Identify the runtime error.
2. Modify the program to handle a missing patient name.
3. Handle invalid age input.
4. Use try-catch-finally appropriately.
5. The program must always display "Registration completed".
6. Submit the COMPLETE corrected Java program.

Faulty Program:

```
import java.util.Scanner;

class PatientRegistration {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

10 marks

5/5 answered

```
System.out.print("Enter patient name: ");
String inputName = sc.nextLine();

String name;

if (inputName.equals("NULL")) {
    name = null;
} else {
    name = inputName;
}

System.out.print("Enter age: ");
int age = sc.nextInt();

System.out.println("Patient Name: "
    + name.toUpperCase());

if (age >= 18) {
    System.out.println("Adult patient");
} else {
    System.out.println("Minor patient");
}

System.out.println("Registration completed");
}
```

The corrected program must prevent NullPointerException and handle invalid age input without terminating unexpectedly.

🛠 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;
2 import java.util.InputMismatchException;
3 class PatientRegistration{
4     public static void main(String[] args){
5         Scanner sc=new Scanner(System.in);
6         try{
7             System.out.println("Enter patient name:");
8             String inputName=sc.nextLine();
9             String name;
10            if(inputName.equals("NULL") || inputName.equals("")){
11                name="Unknown";
12            }else{
13                name=inputName;
14            }
15            System.out.println("Enter age:");
16            int age=sc.nextInt();
17            System.out.println("Patient Name:"+name);
18            if(age>=18){
19                System.out.println("Adult patient")
20            }else{
21                System.out.println("Minor patient")
22            }
23        }catch(InputMismatchException e){
24            System.out.println("Invalid age input")
25        }catch(Exception e){
26            System.out.println("Registration complete")
27        }
28        sc.close();
29    }
30 }
```

Test Input

Hema
20

Output

Enter patient name:
Enter age:
Patient Name:HEMA
Adult patient

🕒 Pending manual review

29 }

30 }

✓ Submitted

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